4.3 price discrimination

AEAAD
ACDDD
DDACB
AEDCB

1. A firm with market power engages in price discrimination to

(A) earn a higher profit ✓
(B) increase consumer surplus
(C) decrease deadweight loss
(D) make its demand more elastic
(E) make its demand more inelastic

2. The price of an airline ticket is typically lower if a traveler buys the ticket several weeks before the flight's departure date rather than on the day of departure. This pricing strategy is based on the assumption that

(A) travelers are not aware of how airline prices change across time
(B) travelers do not have alternative modes of transportation
(C) travelers will pay any price to travel as the departure date approaches
(D) the marginal cost of the last few seats on an airplane is higher than that for the first few seats
(E) travelers' demand becomes less elastic as the departure date approaches ✓

3. Which of the following is true for a price-discriminating firm?

(A) The firm charges different prices to different consumers for the same product. ✓
(B) The firm charges different prices for different products.
(C) The firm pays more per unit of labor than it pays per unit of capital.
(D) The firm pays more per unit of capital than it pays per unit of labor.
(E) The firm sells different quantities to its consumers but charges each of them the same price.

Answer A

Correct. Price discrimination occurs when the firm charges different prices to different consumers for the same product provided that there is no difference in cost exists for charging different prices to different consumers.

4. If a firm engages in perfect price discrimination, it charges

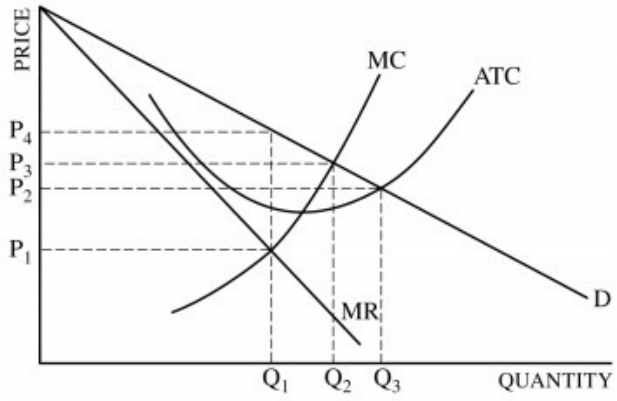
(A) each customer the highest price the customer is willing to pay ✓
(B) each customer the average cost of the product
(C) each customer the lowest price the customer is willing to pay
(D) different prices to customers based on how old they are
(E) different prices to customers based on how many units of output they buy

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Answer A

Correct. Under perfect price discrimination, a firm charges each consumer the maximum price the consumer is willing to pay. By doing so, the firm can increase its profit more than it can by selling the good for a single price.

The following questions refer to the monopoly graph below, where MC = marginal cost, ATC = average total cost, D = demand, and MR = marginal revenue.



5. If the monopolist could engage in perfect price discrimination, the monopolist's total output and the price charged for the last unit of output sold would be

(A) Q_1 and P_1
(B) Q_1 and P_2
(C) Q_1 and P_4
(D) Q_2 and P_3 ✓
(E) Q_3 and P_2

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6. In order for a firm to engage in price discrimi- nation, it must be

(A) able to separate consumers into different groups based on demand elasticities ✓
(B) producing in the inelastic portion of its demand curve to raise its price and increase total revenue
(C) a price taker
(D) experiencing economies of scale in the relevant range of production
(E) experiencing constant marginal cost

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7. Which of the following is necessary for a firm to practice price discrimination?

(A) The government strictly enforces antitrust laws.

(B) The firm is a member of a cartel.

(C) The firm can prevent resale of its goods. ✓

(D) The demand curve for the product is perfectly inelastic.

(E) The demand curve for the product is perfectly elastic.

8. Which of the following is true for a monopolist that engages in perfect price discrimination?

(A) The firm sells the profit-maximizing quantity of the regular monopolist but charges each consumer a price higher than the regular monopoly price.

(B) There is more consumer surplus than exists with a regular monopoly.

(C) The monopolist further restricts output compared to the regular monopoly, creating greater deadweight loss.

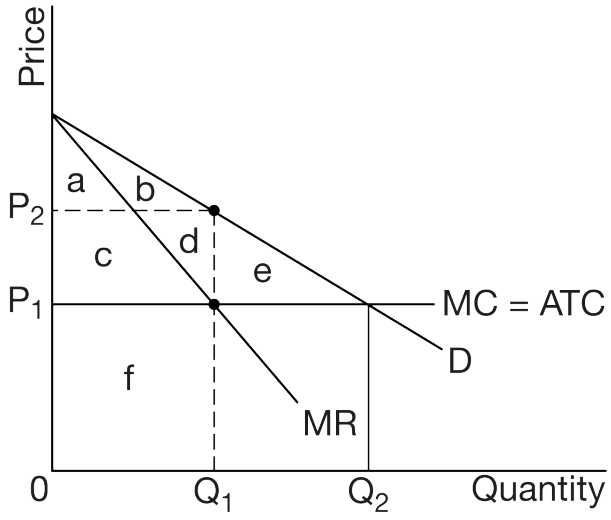
(D) The monopolist sells the allocatively efficient quantity of output. ✓

(E) The monopolist no longer faces a downward-sloping demand curve, becoming a price taker.

Answer D

Correct. The perfectly price-discriminating monopolist sells each unit of output at the maximum price the consumer is willing to pay, as shown by the demand curve. The firm continues to sell output demand equals marginal cost, which is the allocatively efficient output level. Since each consumer pays the maximum price, he or she is willing to pay for the product and there is no consumer surplus. All surplus is producer surplus.

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The graph shows the cost and revenue curves for a profit-maximizing monopolist that produces teddy bears. The letters in the graph represent the enclosed areas.

9. If the monopolist engages in perfect price discrimination, which of the following will happen?

(A) Consumer surplus will remain the same $(a + b)$ and producer surplus will increase by area (e).

(B) Both consumer surplus and producer surplus will decrease by area $(a + b + c + d)$, and the deadweight loss will remain constant.

(C) Consumer surplus will increase by area (e), producer surplus will decrease by area (e), and the deadweight loss will be zero.

(D) Consumer surplus and deadweight loss will be zero because all economic surplus will be transferred to producer surplus. ✓

(E) Both consumer surplus and producer surplus will increase because output produced will increase to Q_2.

Answer D

Correct. Producer surplus will increase, and consumer surplus and the deadweight loss will decrease to zero, because all economic surplus will be captured by the monopolist as profit. Producer surplus will be equal to area $(a + b + c + d + e)$.

10. If the monopolist charges a single price for teddy bears, which of the following describes an accurate outcome?

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(A) The consumer surplus equals area (a), the producer surplus equals area (c), and the deadweight loss equals area $(b + d + e)$.

(B) The consumer surplus equals area (b), the producer surplus equals area $(c + d)$, and the deadweight loss equals area $(d + e)$.

(C) The consumer surplus equals area $(a + b)$, the producer surplus equals area (c), and the deadweight loss equals area $(e + f)$.

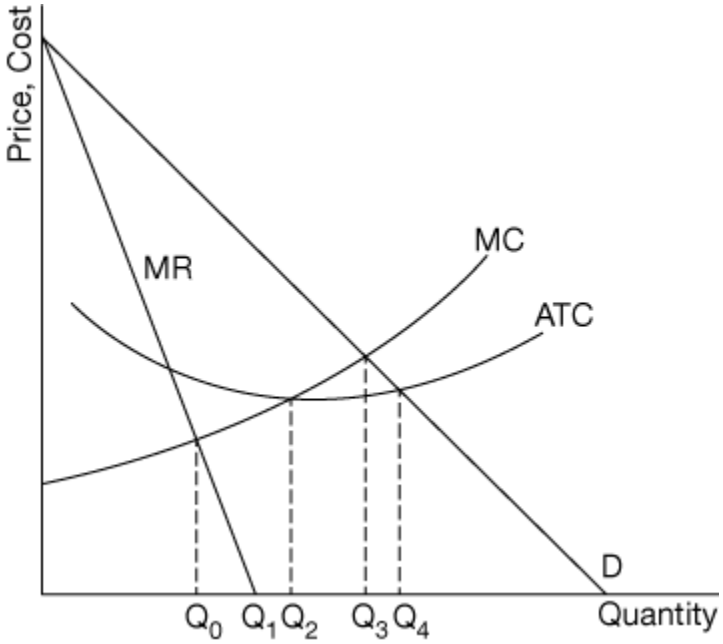
(D) Consumer surplus equals area $(a + b)$, producer surplus equals area $(c + d)$, and deadweight loss equals area (e). ✓

(E) The consumer surplus equals area (a), the producer surplus equals area $(c + f)$, and the deadweight loss equals area $(b + d + e)$.

Answer D

Correct. The consumer surplus is the area under the demand curve and above the price, given by area $(a + b)$; the producer surplus is equal to the economic profit of the monopoly, given by area $(c + d)$; and the deadweight loss is given by area (e), indicating the reduction in total economic surplus.

11.



The graph above shows the demand (D), marginal revenue (MR), marginal cost (MC), and average total cost (ATC) curves for a monopoly. Based on the information in the graph provided, what is the profit-maximizing quantity for a monopolist that engages in perfect price discrimination?

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(A) Q_0
(B) Q_1
(C) Q_2
(D) Q_3 ✓
(E) Q_4

Answer D

Correct. Q_3 is where P (demand line) is equal to MC. This is where the firm is allocatively efficient and practicing perfect price discrimination. With perfect price discrimination, a monopolist produces the quantity where price equals marginal cost (just as a competitive market would) but extracts all economic surplus associated with its product and eliminates all deadweight loss.

12.

Price ($)	Quantity Demanded
1	9
2	8
3	7
4	6
5	5
6	4
7	3
8	2
9	1

A profit-maximizing monopolist faces the market demand schedule provided and has a constant marginal cost of $\$5$. If the monopolist engages in perfect price discrimination, its total revenue will equal which of the following?

(A) $\$10$
(B) $\$15$
(C) $\$25$
(D) $\$35$ ✓
(E) $\$45$

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Answer D

Correct. The firm can charge the maximum possible price for each unit, which enables the firm to capture all available consumer surplus for itself. In perfect price discrimination the firm knows the exact price that the consumer will pay. If $\mathrm{P} = \mathrm{MC}$ at $\$5$, then the firm will sell up to the fifth unit and have a total revenue of $\$5 + \$6 + \$7 + \$8 + \$9 = \$35$.

13. Which of the following enables a seller to capture the entire consumer surplus in a market?

(A) Perfect price discrimination ✓
(B) Perfect competition
(C) An excise tax on buyers
(D) Effective price ceiling
(E) Effective price floor

14. Assume a profit-maximizing monopolist is able to price discriminate, dividing its consumers into two distinct groups charging each a different price. Based on this information, which of the following is true?

(A) The monopolist would earn greater profit charging each consumer the same price, not charging two different prices.
(B) The price discrimination can only be successful if one group can resell the product but the other group cannot.
(C) The group with the more elastic demand will pay the lower price. ✓
(D) The consumers with less willingness to pay will pay the higher price.
(E) The group with greater production costs will pay a lower price.

Answer C

Correct. By charging each group a different price, based on willingness to pay, the producer makes a greater profit. The group with the greater willingness to pay, or less elastic demand, pays a higher price, and the group that has the more elastic demand will pay the lower price.

15. A firm with market power engages in price discrimination in order to

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(A) cover the cost of serving each consumer

(B) increase its profits ✓

(C) charge a lower price

(D) produce the allocatively efficient quantity

(E) increase consumer surplus

Answer B

Correct. A firm sets its price equal to the maximum amount a consumer is willing to pay for that unit to capture additional consumer surplus and to increase its profits.

16. If a monopolist can increase output and profit by engaging in perfect price discrimination, allocative efficiency will

(A) improve even though consumer surplus goes to zero ✓

(B) possibly improve or get worse depending on whether consumer surplus increases or decreases

(C) possibly improve or get worse depending on whether consumer surplus increases by more or less than the increase in producer surplus

(D) get worse because producer surplus increases by more than the increase in consumer surplus

(E) be unchanged because although producer surplus increases, total surplus is unaffected

Answer A

Correct. A monopolist engaging in perfect price discrimination will continue selling output at the maximum price consumers are willing to pay until price is equal to marginal cost (the allocatively efficient outcome). Therefore, the perfectly price-discriminating monopolist will produce the allocatively efficient quantity of output and consumer surplus will decrease to zero.

17. Price discrimination occurs when

(A) the supply of the product is elastic

(B) a product's average cost is greater than its average revenue

(C) a product's average cost is less than its average revenue

(D) differences in a product's price reflect differences in marginal costs

(E) differences in a product's price do not reflect differences in costs of production ✓

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18. A firm with market power sells its product in two markets. The firm faces the same cost curves in both markets but faces a relatively elastic demand in one market for its product and a relatively inelastic demand in the other market for that same product. Which of the following will increase the firm's profits?

(A) The firm charges a lower price in both markets.

(B) The firm charges a higher price in both markets.

(C) The firm charges a price equal to its marginal cost in both markets.

(D) The firm charges a lower price in the market with elastic demand and a higher price in the market with inelastic demand. ✓

(E) The firm charges a lower price in the market with inelastic demand and a higher price in the market with elastic demand.

Answer D

Correct. Because consumers are more responsive to price changes when demand is elastic, the firm will charge a lower price to consumers with elastic demand to incentivize these consumers to buy more of their product (increasing total revenue and, all else equal, increasing profits). Conversely, consumers with relatively inelastic demand are less responsive to price changes; thus, the firm can increase price without reducing unit sales significantly (increasing total revenue and, all else equal, increasing profits).

19. Which of the following is a necessary condition for price discrimination?

(A) Resale of the product or service is possible.

(B) Buyers have identical elasticities of demand.

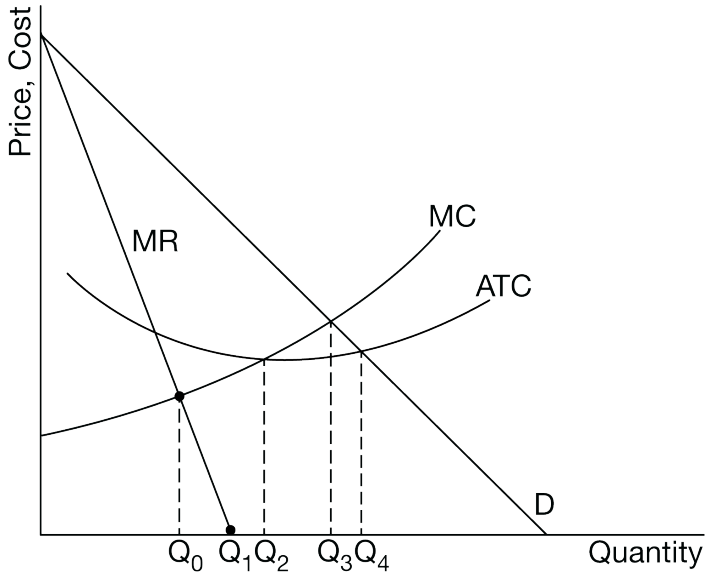
(C) The seller can separate consumers according to their elasticities of demand. ✓

(D) The seller is a perfectly competitive firm.

(E) Buyers are aware of prices charged to other buyers.

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20.



Based on the information in the graph above, what are the profit-maximizing output quantities for a single-price monopolist and for a monopolist that engages in perfect price discrimination?

(A) For a single-price monopolist, Q_0. With perfect price discrimination, Q_1.

(B) For a single-price monopolist, Q_0. With perfect price discrimination, Q_3. ✓

(C) For a single-price monopolist, Q_1. With perfect price discrimination, Q_3.

(D) For a single-price monopolist, Q_2. With perfect price discrimination, Q_4.

(E) For a single-price monopolist, Q_0. With perfect price discrimination, Q_4.

Answer B

Correct. Profit maximization requires the firm to produce where marginal revenue equals marginal cost. For a single-price monopoly, marginal revenue is less than the price. In the graph above, the single-price profit-maximizing firm would produce at Q_0. For a firm that can perfectly price discriminate, the marginal revenue curve and the demand curve are the same curve. Therefore, if the profit-maximizing firm could perfectly price discriminate, it would produce Q_3.